Outlook with Alexa

Amazon has silently updated its Alexa assistant this week to add support for Outlook.com. **http://dgit.in/outlookAlexa**

Lightlines for phone addicts

A city in Netherlands has reportedly been testing "Plus Lightlines" to alert distracted phone users at road crossings. **http://dgit.in/lightlines**

Cartosat-2D undergoing Solar Panel Deployment Test

Utilising Satellite-Automatic Identification System (S-AIS) technology, Spire provides its customers with the ability to track down the location of their shipping vessels in the open seas.

Debuting with ISRO on the other hand are Kazakhstan and the UAE. They have already tasted space success with their satellites Al-Farabi-1 & Nayif-1 respectively. These student built satellites are serving as the test-bed for understanding technologies associated with satellite-based observation and communication. This is their first step towards bigger goals.

The BGUSat (Israel) & PEASSS (the Netherlands) satellites are among the ones validating newer technologies of material, navigation & power-supply for future space sojourns.

The launch

To protect the Nano-satellites from damage during launch turbulence, they were each housed inside a protective casing. A Quadpack CubeSat Dispenser accommodates up to four satellites. Each dispenser was mounted in a carefully calculated angle on the PSLV along with the larger Cartosat-2D and two INS satellites. A protective heat shield also known as the Payload Fairing, covered the satellites.

Upon exiting the Earth's atmosphere, the Fairing parted ways to prepare the satellites. The PSLV first released the Cartosat-2D. The system implemented a combination of measures such as rocket manoeuvring the Stage IV trajectory and orientation of the mounted Quadpack. Later, the Nano-satellites finally emerged out of the housing, safely floating out into space at regular intervals. Within a span of 1,722 seconds from launch, all the satellites were deployed establishing a new space milestone.

The array of 25 Quadpack Dispensers, protecting the 96 Nanosatellites placed within it

Future

India's space scene is witnessing an exciting churning. Encouraged by ISRO's success, private enterprises have begun foraying into this burgeoning sector. The most famous would be Bengaluru-based Axiom Research Labs' TeamIndus. Shortlisted for Google's Lunar XPRIZE, they are preparing to launch their own Moon mission on the PSLV in 2018. Lesser known, but perhaps far more ambitious is Bellatrix Aerospace, that plans to develop its own family of launchers. Together with the others, like Earth2Orbit, Dhruva Space and Applied Research & Development Laboratories (ARDL), a momentum is gradually building up towards the liberalisation of space expertise. Established Indian businesses supplying space sub-systems to ISRO are moving towards expanding scope of their involvement. This ecosystem could inspire a large number of interested graduates to explore this sector and probably reform education with relevant additions.

As ISRO shares its goals and requirements, it would be beneficial to the existing and emerging companies as well in pursuit of achieving bigger goals with national interest. Recently, a reusable launch vehicle – Technology Demonstrator [RLV-TD] concept was flight-tested in an attempt to reduce operating costs in the future. Similar to the erstwhile American Space Shuttle, it would enable ISRO to undertake repeated space missions using the same launch vehicle – tank-up, launch, deploy, land, repeat.

Israeli BGUSat Satellite being put inside a QuadPack CubeSat deployer

ISRO is heading towards developing a full spectrum of indigenous launch capable rockets. It's already spearheaded by the Geosynchronous Satellite Launch Vehicle (GSLV) series and plans are underway to evolve a Unified Launch Vehicle (ULV) & Heavy-lift Launch Vehicle [HLV] concept. Dependability and cost-effectiveness are its USP and eventually India could find itself in an enviable position in the highly monopolised launch industry. Minus Geopolitical machinations, India has the potential to emerge as the numero uno destination for countries looking for some space. It's the country's responsibility to stay focussed, play its cards right to ultimately achieving pure space glory.